Grade 10 Global Economics — Term 2 Review Outline

2026–2027

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1 Introduction

In Lesson 1 (C1–C4), you assign numbers to complete outcomes, group their probabilities, and begin using models for single events and fixed trial counts. In Lesson 2 (C5–C7), the observation rule tells you whether to count trials to a target or count events over a fixed exposure. In Lesson 3 (C8–C9), you interpret expected return and financial risk and decide whether a model fits the evidence.

Use the same routine throughout the term.

1. Define the variable in words and give its units. For a binary trial, name the event called success.
2. Identify what is fixed and what is random.
3. State the support (values with positive probability). Check the model assumptions or validate the supplied probability table.
4. Translate the wording into an event. Mark which endpoints are included.
5. Show the formula and substitution, check the result, and interpret it in context.

Probabilities are unitless. Keep exact expressions or full precision in your working. Round final probabilities to four decimal places unless that would hide a small positive probability.



2 Summarized Outline

Select a criterion to jump to its worked example. The next page gives the event and model tables.

Criterion	Lesson	When asked to...	Method	Show in your working
C1	1	Turn outcomes into a distribution	Map outcomes, then group by value	Outcome masses, assigned values, support, total mass
C2	1	Read or check a PMF or CDF	Accumulate masses or recover jumps	Included values, endpoint heights, validity checks
C3	1	Describe one yes/no event	Build a Bernoulli indicator	Meaning of 0 and 1, both masses, money mapping
C4	1	Count successes in fixed trials	Use binomial point masses, a sum, or a complement	Four conditions, n, p, and arrangement count
C5	2	Wait for the first success	Use geometric point or tail probabilities	Failure–success sequence, cutoff, any conditioning
C6	2	Wait for success r	Fix the final success and count earlier arrangements	r, p, final-trial condition, deadline sum or check
C7	2	Count events over an exposure	Convert units and use Poisson	Exposure, assumptions, λ, included counts
C8	3	Compare expected gains and risk	Compute weighted moments and check downside	Calculation table, units, loss chance and worst value
C9	3	Choose or justify a model	Follow the observation rule and test assumptions	Support, parameters, and why an alternative fails

Criterion map

Reading the markers. Outcome labels, primary variable, second variable, third variable (each variable includes its values and units), unitless probabilities, and quantities to determine.

Quick reference for translating events

Write $p_X(a) = P(X = a)$ for the probability mass function (PMF) and $F_X(a) = P(X \leq a)$ for the cumulative distribution function (CDF). The left limit $F_X(a^-) = P(X < a)$ excludes any mass at a. Use T for a stopping trial and m for a deadline.

Words	**Event**	**Calculation**	**Included values if $a = 2$**
Exactly a	$X = a$	$p_X(a) = F_X(a) - F_X(a^-)$	2
Less than a	$X < a$	$F_X(a^-)$	0
At most a	$X \leq a$	$F_X(a)$	0, 2
At least a	$X \geq a$	$1 - F_X(a^-)$	2, 5
More than a	$X > a$	$1 - F_X(a)$	5
By trial m	$T \leq m$	$F_T(m)$	Includes trial m
After trial m	$T > m$	$1 - F_T(m)$	Starts at trial $m + 1$

The right column uses the **C2** score support $\{0, 2, 5\}$ for the first five rows. With masses $0.2, 0.5, 0.3$, less than 2 has probability 0.2000, while at most 2 has probability 0.7000. A cutoff can fall in a gap. For example, $1 < X < 5$ contains just 2, so its probability is 0.5000.

Quick reference for choosing a model

Here p is the success probability per trial and $q = 1 - p$. The listed supports assume $0 < p < 1$ and $\lambda > 0$. Mass outside the support is zero. Both waiting-time models include the final successful trial.

Model	**Fixed**	**Variable**	**Support**	**Parameters and assumptions**	**Point mass**
Bernoulli	**One observation**	Indicator X	$\{0, 1\}$	p. Two exclusive, exhaustive outcomes.	$p^x q^{1-x}$
Binomial	n **trials**	Success count X	$\{0, \ldots, n\}$	n, p. Independent binary trials, constant p.	$\binom{n}{k} p^k q^{n-k}$
Geometric	**First success** as target	Total trials T	$\{1, 2, \ldots\}$	p. Independent binary trials, constant p.	$q^{t-1} p$
Negative binomial	Target of r **successes**	Total trials T	$\{r, r+1, \ldots\}$	r, p. Independent binary trials, constant p.	$\binom{t-1}{r-1} p^r q^{t-r}$
Poisson	**Exposure** (e.g., time or length)	Event count X	$\{0, 1, \ldots\}$	λ. Constant rate, independent disjoint counts, isolated events.	$e^{-\lambda} \lambda^k / k!$

Here λ = rate × exposure is the mean count in the chosen exposure. The binomial coefficient $\binom{n}{k} = n!/[k!(n-k)!]$ counts ways to choose k positions from n. The factorial $k!$ is the product of the positive integers through k, with $0! = 1$.

3 Step-by-Step Outline

3.1 C1 — From outcomes to a probability mass function

Use this when several complete outcomes can produce the same numerical result, or when a probability table has an unknown mass.

1. List the complete outcomes ω, assign each a value $X(\omega)$, and record its probability. Multiply across independent decisions when the context supplies independence.
2. List the supported values and their units. For each value x, add the probabilities of all outcomes that map to it.
3. Set the sum of the masses equal to 1 to check or solve for unknowns. Check non-negativity and include any tail beyond the displayed values.

Notebook buyers. Three visitors independently buy with probability 0.4 each. B means buy and N means no purchase. In the table, each label records the decisions of all three visitors in order. The markers identify outcome labels, assigned values and units, and unitless probabilities.

Complete outcomes ω	$x = X(\omega)$ (buyers)	Mass of each outcome	Grouped mass $p_X(x)$
NNN	0	$0.6^3 = 0.216$	0.2160
BNN, NBN, NNB	1	$0.4(0.6)^2 = 0.144$	$3(0.144) = 0.4320$
BBN, BNB, NBB	2	$(0.4)^2(0.6) = 0.096$	$3(0.096) = 0.2880$
BBB	3	$0.4^3 = 0.064$	0.0640

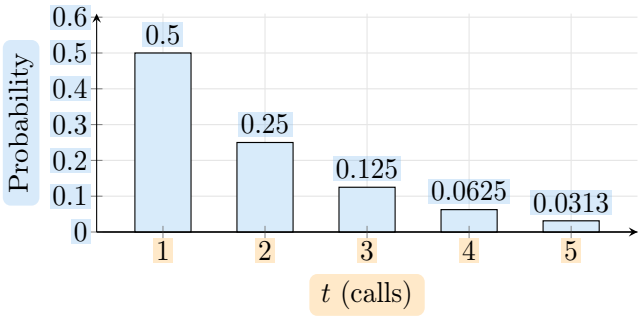
The support is $\{0, 1, 2, 3\}$ buyers. For instance, BNN and NBN describe different buyers but both contribute to $p_X(1)$. Exactly one buyer has probability 0.4320. If the last mass were missing, $p_X(3) = 1 - (0.216 + 0.432 + 0.288) = 0.0640$, the chance all three buy.

Normalizing weights. If refunds of $0, 2, 5$ USD have masses $c, 2c, 3c$, then $6c = 1$, so $c = 1/6$. The resulting masses $1/6, 1/3, 1/2$ are positive and total 1. A refund of at least 2 USD has probability $1/3 + 1/2 = 5/6 \approx 0.8333$.

Infinite support. If each independent call sells with probability $1/2$, the first sale on call t has mass $p_T(t) = 2^{-t}$ for $t = 1, 2, \ldots$. The first m terms sum to $1 - 2^{-m}$, which tends to 1. Beyond call 5 the omitted mass is $P(T > 5) = 2^{-5} = 1/32 = 0.03125$.



Buyer-count PMF. Each height is the grouped probability in the last column above.



First-sale PMF, first five calls. The bars total 31/32. Calls $6, 7, \ldots$ carry the remaining 1/32.

3.2 C2 — Read, build, and verify a CDF

Use this when you are given individual masses or a cumulative table or graph, especially when the question changes an endpoint.

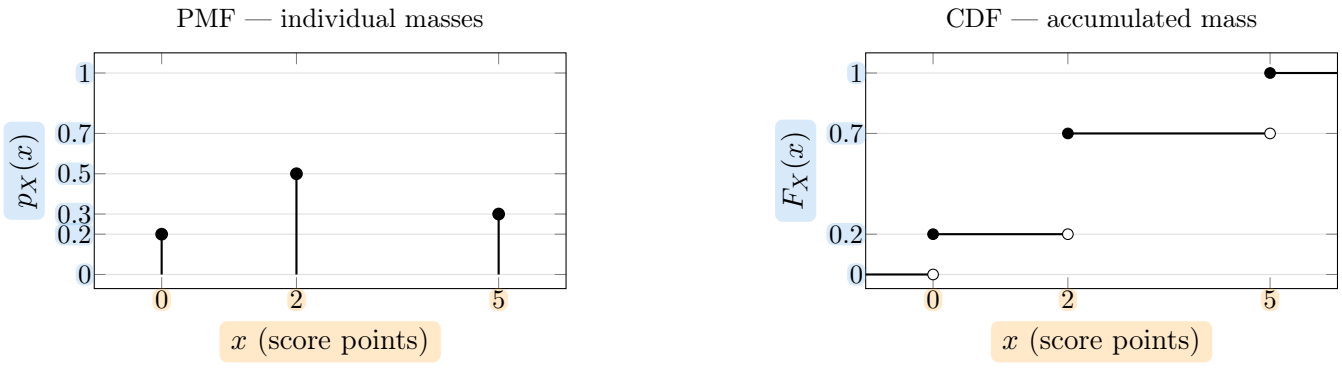
1. Verify that every PMF mass is non-negative and the total is 1.
2. Build $F_X(\,x\,) = \sum_{u \le x} p_X(u)$ by accumulating the masses in order. Keep the height constant between supported values.
3. To go back to the PMF, subtract the height just before a jump from the height at the jump. Check that the CDF never decreases, is right-continuous, and tends to 0 on the far left and 1 on the far right.
4. Mark the included values. Add their masses, subtract CDF heights, or use a complement.

Scores on irregular support. X has support $\{0, 2, 5\}$ in score points. Its masses are positive and $0.2 + 0.5 + 0.3 = 1$. For any real cutoff x,

x	0	2	5
$p_X(x)$	0.2	0.5	0.3
$F_X(x)$	0.2	0.7	1

$$F_X(x) = \begin{cases} 0 & x < 0, \\ 0.2 & 0 \le x < 2, \\ 0.2 + 0.5 = 0.7 & 2 \le x < 5, \\ 0.7 + 0.3 = 1 & x \ge 5. \end{cases}$$

Reading the jumps back gives $p_X(0) = 0.2 - 0 = 0.2000$, $p_X(2) = 0.7 - 0.2 = 0.5000$, and $p_X(5) = 1 - 0.7 = 0.3000$.



The filled dot gives the CDF value at each jump. The open dot gives its left limit. The two outer steps continue horizontally.

$$P(0 < X \le 5) = F_X(5) - F_X(0) = 1 - 0.2 = 0.8000,$$
$$P(X < 5) = F_X(5^-) = 0.7000, \qquad P(X \le 5) = F_X(5) = 1.0000,$$
$$P(X > 2) = 1 - F_X(2) = 1 - 0.7 = 0.3000.$$

For $P(a \le X \le b)$, subtract $F_X(a^-)$ from $F_X(b)$ to retain the mass at a. Here a strict cutoff at 5 excludes the 0.3000 chance of scoring exactly 5.

Reject invalid candidates. Masses $0.6, 0.5, -0.1$ total 1 but contain a negative mass. Masses $0.2, 0.3, 0.3$ total only 0.8. Reject a CDF that steps down, uses the open-dot height as its value at a jump, or levels off at 0.9 forever on the right.

3.3 C3 — Bernoulli indicators

Use this when one observation has **two mutually exclusive outcomes** that cover all possibilities, such as default within one year or no default within one year.

1. Define the event A and set $X = 1$ if A occurs and $X = 0$ otherwise. The indicator is unitless.
2. Show $p_X(1) = p$ and $p_X(0) = 1 - p$. To indicate the opposite event, set $Z = 1 - X$ and swap the two masses.
3. Define a separate monetary variable if needed. Match each monetary outcome with its indicator value.

One loan default. A loan has one-year default probability $p = 0.04$. Let $X = 1$ for default, so the support is $\{0, 1\}$. A default costs 800 USD, giving loss $L = 800X$ in USD.

Outcome	X	$Z = 1 - X$	L (USD)	Probability
Default	1	0	800	$p = 0.0400$
No default	0	1	0	$1 - 0.04 = 0.9600$

Thus $P(Z = 1) = 0.9600$ and $P(L = 800) = 0.0400$. The word *success* names the event coded 1, which here triggers a loss.



3.4 C4 — Binomial success counts

Use this when you **count successes** in a **fixed number of trials**.

1. Verify all four conditions by checking fixed n, two outcomes per trial, independence, and constant success probability p. Define X as the success count with support $\{0, \ldots, n\}$ for $0 < p < 1$.
2. For exactly k, multiply one arrangement's probability $p^k(1-p)^{n-k}$ by $\binom{n}{k}$, the number of ways to choose the k success positions.
3. For a range, add one mass per included count. For at least one, subtract the zero-success mass from 1.

Five independent loans. Observe all five for one year. Each defaults with probability 0.1, so $X \sim \text{Bin}(5, 0.1)$ counts defaults. Exactly one default can occur in any of five positions.

$$P(X = 1) = \binom{5}{1}(0.1)(0.9)^4 = 0.32805 \approx 0.3281,$$
$$P(X \leq 1) = P(X = 0) + P(X = 1) = (0.9)^5 + 5(0.1)(0.9)^4 \approx 0.9185,$$
$$P(X \geq 1) = 1 - P(X = 0) = 1 - (0.9)^5 \approx 0.4095.$$

About 40.95% of such portfolios would have one or more defaults under these assumptions. If a market shock makes the loans default together, a default on one loan changes what you know about the others. The independence calculation then needs a different joint model (see **C9**).



3.5 C5 — Geometric waiting to the first success

Use this when independent binary trials with constant success probability p continue **until the first success**.

1. Define T as total trials through the first success, with support $\{1, 2, \ldots\}$ for $0 < p < 1$. Set $q = 1 - p$.
2. For *on trial* t, write $t - 1$ failures followed by one success, giving $q^{t-1}p$.
3. For *after trial* m, all first m trials fail, giving q^m. For *by trial* m, use $1 - q^m$. Convert money or failures to a total-trial cutoff first.

A window and a changed starting point. Suppose independent calls sell with $p = 0.3$. First sale on call 3 is FFS, so $P(T = 3) = (0.7)^2(0.3) = 0.1470$. After call 3 has probability $P(T > 3) = (0.7)^3 = 0.3430$, and by call 3 has probability $P(T \le 3) = 1 - (0.7)^3 = 0.6570$.

$$P(2 < T \le 5) = P(T > 2) - P(T > 5) = 0.7^2 - 0.7^5 \approx 0.3219,$$
$$P(T \le 5 \mid T > 2) = \frac{0.7^2 - 0.7^5}{0.7^2} = 1 - 0.7^3 = 0.6570.$$

From the start, the window requires two initial failures. Given those failures, the next three calls retain their original chances. With $F = T - 1$ failures before the sale, $F = 2$ is the event $T = 3$.

Budget variant. At 3 USD per call, a budget of 12 USD buys at most $\lfloor 12/3 \rfloor = 4$ calls. With the budget example's $p = 0.25$, the chance of making the first sale within budget is $P(T \le 4) = 1 - (0.75)^4 \approx 0.6836$.



3.6 C6 — Negative binomial waiting to the target success

Use this when independent binary trials with constant p continue **until a fixed target of r successes**.

1. Define T as total trials through success r. Its support starts at r. For $T = t$, fix trial t as a success.
2. Choose $r - 1$ success positions among the first $t - 1$ trials. Multiply the resulting $\binom{t-1}{r-1}$ arrangements by $p^r(1-p)^{t-r}$.
3. For a deadline m, sum the masses from $t = r$ to m. Check by counting at least r successes in a fixed window of m trials. If $m < r$, the deadline probability is zero.

Third sale with $p = 0.4$. T counts calls, with support $\{3, 4, \ldots\}$. The third sale on call 5 requires two sales among the first four calls and a sale on call 5. Thus $P(T = 5) = \binom{4}{2}(0.4)^3(0.6)^2 = 0.13824 \approx 0.1382$.

$$P(T \le 6) = (0.4)^3[1 + 3(0.6) + 6(0.6)^2 + 10(0.6)^3] = 0.45568 \approx 0.4557,$$
$$P(B \ge 3) = 1 - \sum_{j=0}^{2} \binom{6}{j}(0.4)^j(0.6)^{6-j} = 1 - 0.54432 = 0.45568, \quad B \sim \mathrm{Bin}(6, 0.4).$$

Both calculations give the chance of reaching the target within six calls. In the binomial check, imagine **observing all six calls** even if the target is reached early. Failures satisfy $F = T - r$, so $T = 5$ gives $F = 2$. With $r = 1$, the coefficient becomes $\binom{t-1}{0} = 1$ and the PMF becomes geometric.



3.7 C7 — Poisson counts over fixed exposure

Use this when you **count events** over a **fixed time, length, area, or other exposure**.

1. Define the count X and exposure. Check a **constant rate**, **independent counts in disjoint intervals**, and **isolated events** (occurring singly).
2. Convert the rate and exposure to compatible units. Multiply to get the mean count λ. The support is $\{0, 1, 2, \ldots\}$ for $\lambda > 0$.
3. Use $e^{-\lambda}\lambda^k/k!$ for exactly k. Sum through k for at most k, or subtract the sum through $k-1$ from 1 for at least k.

Arrivals over 150 minutes. Assume isolated arrivals at a constant rate of 0.8 per hour and independent counts in disjoint intervals. Convert 150 minutes to $150/60 = 2.5$ hours, so $\lambda = (0.8 \text{ arrivals/hour})(2.5 \text{ hours}) = 2$ arrivals.

$$P(X = 2) = e^{-2}\frac{2^2}{2!} = 2e^{-2} \approx 0.2707,$$
$$P(X \le 2) = e^{-2}\left(1 + 2 + \frac{2^2}{2!}\right) = 5e^{-2} \approx 0.6767,$$
$$P(X \ge 3) = 1 - 5e^{-2} \approx 0.3233.$$

About 32.33% of such intervals would contain at least three arrivals.

Cost threshold with the same mean. Suppose claims arrive under the Poisson assumptions at 0.4 claims per day and each costs 100 USD. Over five days the mean is again $\lambda = 0.4(5) = 2$ claims. If $L = 100X$ is total cost in USD, then $L \ge 300$ means $X \ge 3$. The same complement gives $P(L \ge 300) = 1 - 5e^{-2} \approx 0.3233$. Show the conversion from the money threshold to the count before using the PMF.

When events cluster. If a market shock triggers many defaults together, the isolated-event assumption fails for that default count. To calculate a tail probability, you need information about the frequency and size of the clusters. See **C9** for how a shared shock can change a portfolio probability.

3.8 C8 — Expected value, variance, and financial risk

Use this when a PMF gives uncertain gains, costs, or counts and you need to summarize the average and the spread.

1. Validate the PMF. For each supported value, show x, $p_X(x)$, $xp_X(x)$, and $x^2p_X(x)$.
2. Sum the third column for the mean $\mu = E(X)$ and the fourth for $E(X^2)$. Subtract μ^2 to get variance, then take its square root for standard deviation.
3. For $Y = aX + b$, use $E(Y) = aE(X) + b$, $\mathrm{Var}(Y) = a^2\,\mathrm{Var}(X)$, and $\sigma_Y = |a|\sigma_X$. Compare loss probability and worst outcome as well as the spread.

Claims and a reserve balance. X counts daily claims. A firm starts with 100 USD and pays 30 USD per claim, so $Y = 100 - 30X$ is the remaining reserve in USD.

x (claims)	$p_X(x)$	$xp_X(x)$ (claims)	$x^2p_X(x)$ (claims^2)
0	0.6	0	0
1	0.3	$1(0.3) = 0.3$	$1^2(0.3) = 0.3$
2	0.1	$2(0.1) = 0.2$	$2^2(0.1) = 0.4$
Total	1	0.5	0.7

Variance is the weighted average of squared deviations from the mean. Expanding the square and using $\sum_x p_X(x) = 1$ gives the shortcut:

$$\sum_x (x-\mu)^2 p_X(x) = E(X^2) - 2\mu E(X) + \mu^2 = E(X^2) - \mu^2.$$

The mean is 0.5 claims per day. The variance is $0.7 - 0.5^2 = 0.45$ claims^2 and $\sigma_X = \sqrt{0.45} \approx 0.6708$ claims. For the reserve,

$$E(Y) = 100 - 30(0.5) = 85 \text{ USD},$$
$$\mathrm{Var}(Y) = (-30)^2(0.45) = 405 \text{ USD}^2, \qquad \sigma_Y = \sqrt{405} \approx 20.1246 \text{ USD}.$$

Over many comparable days the average remaining reserve approaches 85 USD. On a single day the supported balances are 100, 70, and 40 USD.

Equal variance can hide different losses. Investment A gives -10 or 10 USD with probability 0.5 each. B gives $-20, 0, 20$ USD with masses $0.125, 0.75, 0.125$. The opposite gains cancel, giving both means 0 USD. Their second moments are $2(10^2)(0.5) = 100$ and $2(20^2)(0.125) = 100$ USD^2, so both variances are 100 USD^2.

	Investment A	**Investment B**
Probability of a loss	0.5000	0.1250
Worst gain (USD)	-10	-20

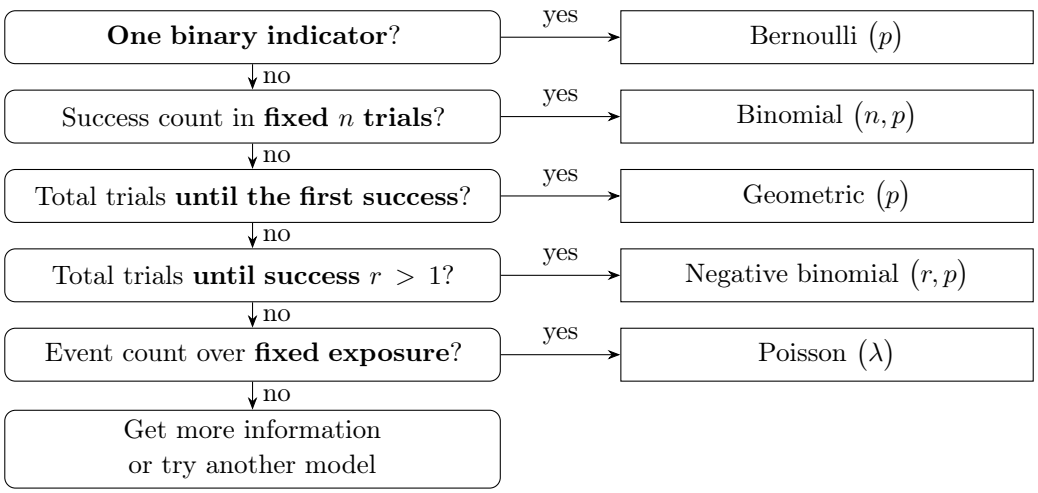
B loses less often, but its worst loss is twice as large.



3.9 C9 — Select and justify the probability model

Use this when several models sound plausible or the question asks whether enough information is available.

1. Define **what the variable measures** and **what the observation rule fixes**.
2. Follow the chart to a candidate. State its support, units, parameters, and assumptions. Explain which feature rules out a plausible competing model.
3. Calculate with the justified model. If a required assumption is unresolved, show which probability depends on the missing information.



A yes branch gives a candidate. Verify its assumptions using the model table.

On-time invoices. Observe all five invoices. Each is independently paid on time with probability 0.8. X counts on-time payments, with support $\{0, \ldots, 5\}$ payments. A **fixed** $n = 5$, **binary outcomes**, **independence**, and **constant** $p = 0.8$ justify binomial. Geometric would measure the trials **to the first on-time payment**.

$$P(X = 4) = \binom{5}{4}(0.8)^4(0.2) = 5(0.4096)(0.2) = 0.4096.$$

This is the chance that exactly one invoice is late.

Shared default shock. Each of five loans has one-year default probability 0.1, so one loan's indicator has a Bernoulli PMF. The count D is between 0 and 5 defaults. Its support and aggregate probabilities depend on **how the loans default together**.

- If defaults are **independent**, $D \sim \text{Bin}(5, 0.1)$ and $P(D = 5) = (0.1)^5 = 0.00001$.
- If a shock with probability 0.1 causes **all five to default together** and otherwise none default, the support is $\{0, 5\}$ and $P(D = 5) = 0.1000$.

Both cases give each loan the same default probability. To assign a chance to all five defaulting, you need a joint model for the portfolio.